

Lab 1a Linux Commands – Prerequisite for Hadoop Jobs and HDFS Commands

1) “Find the command that tells you exactly where you are in the directory structure.”

- How would you show the full path to the directory you’re in?

PWD → **P**rint **W**orking **D**irectory

Ready to start your first linux lab?

Let's begin! Type this and press Enter below and see what happens:

```
echo "Hello LabEx"
```

```
labex:project/ $ ls
labex:project/ $ pwd
/home/labex/project
labex:project/ $
```

- What happens if you run this after navigating into several nested folders?

- It will show all the folders which we created and gives path of it.

```
labex:project/ $ ls
labex:project/ $ pwd
/home/labex/project
labex:project/ $ mkdir
mkdir: missing operand
Try 'mkdir --help' for more information.
labex:project/ $ mkdir MTech_Sangram_Lembe
labex:project/ $ cd MTech_Sangram_Lembe
labex:MTech_Sangram_Lembe/ $ mkdir labx
labex:MTech_Sangram_Lembe/ $ cd labx
labex:labx/ $ pwd
/home/labex/project/MTech_Sangram_Lembe/labx
labex:labx/ $
```

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2) “Find the command to create a new folder.”

What it stands for: Make Kirectory.

Basic Usage: `mkdir folder_name`

- How do you make nested folders in one go (`mkdir -p path/to/dir`)?

To create a path with multiple levels of folders at once (like a folder inside a folder inside another folder), use the `-p` (parents) flag.

- **The Command:** `mkdir -p path/to/dir`

```
labex:labx/ $ cd ..
labex:MTech_Sangram_Lembe/ $ mkdir -p project/data/raw
labex:MTech_Sangram_Lembe/ $ cd project/data
labex:data/ $ pwd
/home/labex/project/MTech_Sangram_Lembe/project/data
labex:data/ $
```

- How do you create folders whose names contain spaces?

In Linux, spaces are used to separate different commands or arguments. If you type `mkdir My New Folder`, Linux will think you want to create three separate folders: one named `My`, one named `New`, and one named `Folder`.

Option A: Use Quotes

```
labex:project/ $ cd ..
labex:MTech_Sangram_Lembe/ $ mkdir "My New Folder"
labex:MTech_Sangram_Lembe/ $ pwd
/home/labex/project/MTech_Sangram_Lembe
labex:MTech_Sangram_Lembe/ $ cd "My New Folder"
labex:My New Folder/ $ pwd
/home/labex/project/MTech_Sangram_Lembe/My New Folder
labex:My New Folder/ $
```

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Option B: Use the Backslash (**Escape Character**) Put a backslash \ before every space.

```
labex:project/ $ mkdir my\ new\ folder
labex:project/ $ cd my new folder
cd: too many arguments
labex:project/ $ cd "my new folder"
labex:my new folder/ $ pwd
/home/labex/project/my new folder
labex:my new folder/ $
```

3) “Find the command to list all the files and subdirectories in a folder.”

– The basic command to list files and subdirectories is **ls**

```
$ cd ..
(no output)
$ ls
Home_work
Mtech
sangram
```

• How would you list all including hidden files?

```
$ ls -a
.
..
.bash_logout
.bashrc
.profile
Home_work
Mtech
sangram
```

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- How would you list files with detailed info like size and permissions?

To see the "long" version of the list, which includes file permissions, owner, size, and date modified, use the `-l` flag. → `ls -l`

- **Human-Readable:** Use `ls -lh`. The `h` makes the file sizes readable (e.g., **2GB** instead of **2147483648 bytes**).

```
$ ls -l
total 0

$ ls -lh
total 0
```

- How would you list recursively through subfolders?

To see the contents of the current folder **and** every folder inside it, use the capital `-R` flag.
→ `ls -R`

```
/home/user
$ mkdir sangram
(no output)
$ ls -R
.:

```

4) "Find the command to move into another folder."

– move into another folder is `cd`

- How do you go up one level?

To move "backwards" to the parent folder, use the double-dot shortcut:

- 1) **For 1 level up** : `cd ..`
- 2) **For 2 level up**: `cd ../../`

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```
$ ls
Mtech
lembe
sangram
$ cd sangram
(no output)
$ cd ..
(no output)
```

- How do you go back to home quickly?

Type just `cd` and hit Enter.

```
$ mkdir -p jay/mata/di
(no output)
$ cd
(no output)
$ pwd
/home/user
```

- How do you move to a folder by absolute vs relative path?

Feature	Absolute Path	Relative Path
Starting Point	The Root of the whole system (/).	Your Current folder.
Starts with...	A forward slash /.	A folder name or ...

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Example	<code>cd /home/labex/project/data</code>	<code>cd project/data</code>
When to use?	When you want to go to a specific place regardless of where you are now.	When you want to move to a folder that is "near" your current location.

5) “Find the command to remove an empty folder.”

Command: `rmdir folder_name`

```
labex:data/ $ pwd
/home/labex/project/sangram/Mtech/data
labex:data/ $ cd ..
labex:Mtech/ $ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
labex:Mtech/ $ rmdir data
labex:Mtech/ $ pwd
/home/labex/project/sangram/Mtech
```

• What if the folder is not empty — what command should you then use?

`rm -r folder_name`

-r means —> Recursive.

This tells Linux to go inside the folder, delete every file, go into every subfolder, delete those, and finally delete the main folder itself.

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```
/home/labex/project/sangram/Mtech/files/data
labex:data/ $ cd ../..
labex:Mtech/ $ rmdir -r files
rmdir: invalid option -- 'r'
Try 'rmdir --help' for more information.
labex:Mtech/ $ rm -r files
labex:Mtech/ $ pwd
/home/labex/project/sangram/Mtech
```

6) “Find the command to delete a file or folder.”

Command: `rm`

```
labex:Mtech/ $ ls
data.txt  hello.txt
labex:Mtech/ $ rm data.txt
labex:Mtech/ $ ls
hello.txt
labex:Mtech/ $
```

- How would you delete a directory and all its contents in one command? (`rm -r`)

To delete a single file: `rm filename.txt`

To Delete Recursive: `rm -r folder_name`

```
/home/labex/project/sangram/Mtech/files/data
labex:data/ $ cd ../..
labex:Mtech/ $ rmdir -r files
rmdir: invalid option -- 'r'
Try 'rmdir --help' for more information.
labex:Mtech/ $ rm -r files
labex:Mtech/ $ pwd
/home/labex/project/sangram/Mtech
```

- How do you ask for confirmation before deleting? (`rm -i`)
(interactive)

Command: `rm -i filename` or `rm -ri folder_name`

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7) “Find the command to create a new, empty file.”

`touch newfile.txt`

```
labex:Mtech/ $ touch data.txt
labex:Mtech/ $ echo "Hello World" > hello.txt
labex:Mtech/ $ ls
data.txt  hello.txt
```

- Sub: How would you update the timestamp of an existing file?

Scenario	Command	Result
File doesn't exist	<code>touch file.txt</code>	Creates a new empty file.
File already exists	<code>touch file.txt</code>	Updates "Last Modified" time to now .
Specific Timestamp	<code>touch -t 202301010101 file.txt</code>	Sets the timestamp to a specific past date.

Timestamp

```
labex:Mtech/ $ touch data.txt
labex:Mtech/ $ ls -l data.txt
-rw-rw-r-- 1 labex labex 0 Jan 10 16:51 data.txt
labex:Mtech/ $ touch dat.txt
labex:Mtech/ $ touch data.txt
labex:Mtech/ $ ls -l data.txt
-rw-rw-r-- 1 labex labex 0 Jan 10 16:54 data.txt
```


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8) "Find the command to copy a file from one place to another."

Command: cp

- **Basic Syntax:** `cp [options] source destination`
- `cp model.pth ./backups/` (Copies the file model.pth to the backups folder).
- How do you copy entire directories recursively? (`cp -r`)

```
labex:data/ $ cd ..  
labex:Mtech/ $ cp -r data ./sangram/
```

If the `destination_directory` **doesn't exist**, it creates a new folder with that name and puts the contents inside.

If it **already exists**, `cp` will place the `source_directory` **inside** the destination folder

- How would you copy and preserve file attributes (permissions/timestamps)?

Option A: The Preserve Flag (-p) Use this to keep the "mode, ownership, and timestamps."

```
labex:Mtech/ $ cp -p hello.txt dat.txt  
labex:Mtech/ $
```

Option B: The Archive Flag (-a)

Archive and is a shortcut for "recursive + preserve everything."

```
labex:Mtech/ $ cp -a data Mtech
```

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9) "Find the command to move or rename a file."

The command to move or rename is `mv`

- How would you use the same command to rename a file?

```
labex:Mtech/ $ mv dat.txt empty_file.txt
labex:Mtech/ $
```

- How would you move multiple files into one target directory?

```
labex:Mtech/ $ mv dat.txt empt_data.txt data/
```

10) "Find the command to show a file's contents on the screen."

common command to show a file's contents on the screen is `cat`
`cat filename.txt`

- How do you combine two files into one?

Because `cat` is designed to "concatenate" (link things together), you can use it with the redirection operator (`>`) to merge files.

- The Command: `cat file1.txt file2.txt > combined.txt`

```
labex:data/ $ ls
My_file.txt  dat.txt  empt_data.txt
labex:data/ $ cat dat.txt My_file.txt > combined.txt
labex:data/ $ echo "Hii how are you" > combined.txt
labex:data/ $
```

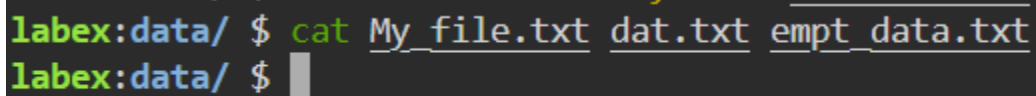
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- What happens if you cat multiple files in one command?

If you run `cat file1.txt file2.txt file3.txt` **without** a redirection operator:

1. Read `file1` and print it to the screen.
2. Immediately after the last line of `file1`, it prints `file2`.
3. Then it prints `file3`.

It looks like one long continuous document in your terminal.



```
labex:data/ $ cat My_file.txt dat.txt empty_data.txt
labex:data/ $
```

11) “Find the command to read help/documentation about another command.”

The command to read help/documentation is `man` → `manual`

- How would you search within the manual page?

`man ls` will open a detailed manual for the list command.

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```
NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default). Sort entries
    alphabetically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file
```

- What is the shortcut to quit the man page viewer?

q key.

```
    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file

    -b, --escape
        print C-style escapes for nongraphic characters

    --block-size=SIZE
        with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M';
        see SIZE format below

    -B, --ignore-backups
        do not list implied entries ending with ~

    -c
        with -lt: sort by, and show, ctime (time of last modification of file sta-
        tus information); with -l: show ctime and sort by name; otherwise: sort
```

Manual page ls(1) line 1 (press h for help or q to quit)



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12) "Find how to quickly get a usage summary for any command."

- How would you use this to learn about options for ls?

use the `--help` flag.

```
ls --help
```

13) "Find the command to print text or variables to the screen."

- How do you redirect that text to a file?

Command: `echo "Model training started" > log.txt`

- How do you append instead of overwrite?

Command: `echo "Model training finished" >> log.txt`

```
labex:Mtech/ $ echo "hi how are you " > hehe.txt
labex:Mtech/ $
```

14) "Find the command to change who can read/write/execute a file."

- How do you make a script executable?

r (read): Can you see the content?

w (write): Can you change or delete it?

x (execute): Can you run it like a program/script?

```
labex:Mtech/ $ echo "echo Hello" > test.sh
labex:Mtech/ $ ./test.sh
zsh: permission denied: ./test.sh
labex:Mtech/ $ chmod +x test.sh
labex:Mtech/ $ ./test.sh
Hello
labex:Mtech/ $
```

1. Symbolic Mode (The "Human" Way)

Symbolic mode is great when you only want to change **one specific thing** without worrying about the other permissions.

Structure: [Who] [Action] [Permission]

- **Who:** u (user), g (group), o (others), a (all).
- **Action:** + (add), - (remove), = (set exactly).
- **Permission:** r (read), w (write), x (execute).

```
labex:Mtech/ $ chmod g+w data.txt
labex:Mtech/ $
```

2. Numeric Mode (The "Math" Way)

Numeric mode is the favorite of MLOps engineers because it is fast and sets the **entire** permission state for everyone (user, group, and others) in one single command.

Read (r) = 4

Write (w) = 2

Execute (x) = 1

No Permission = 0

```
labex:Mtech/ $ chmod 755 test.sh
labex:Mtech/ $
```

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15) "Find the command that lets you run as administrator."

The command that lets you run another command as an administrator is **sudo** (short for **SuperUser DO**).

- What happens if you run a command without sudo when you need permissions?

If you try to perform a restricted action (like editing a system file in `/etc/` or installing software) without **sudo** &

Linux will stop you and display:

Permission denied > or E: Could not open lock file - open (13: Permission denied)

Linux does this as a safety feature. It acts as a "guard" to prevent unauthorized changes to the core operating system.

- How do you edit a system file with elevated privileges?

sudo nano /path/to/system_file

```
labex:Mtech/ $ mkdir /test_folder
mkdir: cannot create directory '/test_folder': Permission denied
labex:Mtech/ $ sudo mkdir /tets_folder
labex:Mtech/ $ ls /
bin                dev                lib32              media              root              srv                usr
boot              etc                lib64              mnt               run               sys                var
code-server_4.18.0_amd64.deb home              libx32             opt               sbin              tets_folder
data              lib                mariadb_repo_setup proc               snap              tmp
```

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16) "Find the command to search for text inside files."

Command: `grep`

The command to search for text inside files is `grep` (short for **G**lobal **R**egular **E**xpression **P**rint).

- **Basic Usage:** `grep "search_term" filename`

```
labex:Mtech/ $ grep "search_term" hello.txt
labex:Mtech/ $
```

- How would you search recursively in a directory?

`grep -r`

```
labex:sangram/ $ grep -r "my_variable" ./my_project
```

- How do you make the search case insensitive?

`grep -i`

17) "Find the command to locate files by name or criteria in a directory tree."

```
labex:sangram/ $ cd Mtech
labex:Mtech/ $ ls
data data.txt hehe.txt hello.txt test.sh
labex:Mtech/ $ find data
data
labex:Mtech/ $ find hello.txt
hello.txt
```

- How would you find files modified within the last 7 days?

Command: `find . -mtime -7`

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```
labex:Mtech/ $ find . -mtime -7
.
./data
./data.txt
./hello.txt
./hehe.txt
./test.sh
```

- How would you execute a command against each found file?

Command: `find . -name "*.log" -exec rm {} \;`

```
labex:Mtech/ $ find . -name "*.log" -exec rm {} \;
labex:Mtech/ $
```

18) "Find the command to archive files into one package."

Command: `tar`

- How do you extract a tar archive?

To "unpack" the files, you replace the `c` (create) flag with the `x` (extract) flag.

- **The Command:** `tar -xvf archive_name.tar`

- How do you list contents without extracting?

Command: `tar -tvf archive_name.tar`

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19) "Find the command to monitor processes and system resource usage."

- How do you quit the monitor?

Unlike most commands that run and then return you to the prompt, `top` stays open and updates every few seconds.

- **The Shortcut:** Press the `q` key
- What key sorts by memory instead of CPU?

By default, `top` sorts processes by %CPU (the ones using the most processing power are at the top). To see which processes are "eating" your RAM (Memory):

- **The Shortcut:** Press `M` (specifically `Shift + m`).
- **To switch back to CPU sorting:** Press `P` (specifically `Shift + p`)

20) "Find the command to show disk usage and space left on drives."

Command: `df`

- How do you view in human-readable format (e.g., MB/GB)?

By default, `df` outputs sizes in 1-kilobyte blocks, which is very hard for humans to read (e.g., `49254308`). To see sizes in KB, MB, or GB, use the `-h` flag.

- **The Command:** `df -h`
- How do you check the size of a specific directory?

While `df` tells you about the whole "drive," the `du` (short for disk usage) command tells you how much space a specific folder or file is taking up.

- **The Command:** `du -sh [directory_name]`

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